Individual and Collective Agency in Teams: Distributed Minds

Executive Summary

A team's intelligence is not the sum of its members' intelligence — it is an emergent property of how individuals coordinate their inference. Under Active Inference, collective agency arises when individual agents align their generative models, distribute the inference task across specialized roles, and develop coordination mechanisms that produce behavior no individual could achieve alone. This module examines how individual agency gives rise to collective agency, the conditions under which groups outperform individuals, and the pathologies that destroy collective intelligence.

Learning Objectives

1. Distinguish **individual agency** from **collective agency** in teams
2. Explain how **distributed cognition** enables teams to solve problems beyond individual capacity
3. Understand the concept of **role specialization** as inference task distribution
4. Identify the conditions under which **groups outperform individuals** (and when they don't)
5. Diagnose pathologies of collective agency: social loafing, groupthink, and coordination failure

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Key Concepts

1. From Individual to Collective Agent

Level	Agent	What It Infers	What It Predicts	What It Does
Individual	Person	My task, my relationships	What happens if I do X	My work, my communication
Dyad	Pair	Shared task state	How my partner will respond	Coordinated paired action
Team	Work group	Project state, stakeholder state	Delivery timeline, risk	Collective execution

Collective agency is not automatic. A group becomes a collective agent only when:

- Members develop **shared representations** (aligned generative models)
- The group develops **emergent capabilities** not present in any individual
- There is **coordination** that is not centrally controlled but emerges from interaction

Case Study — Surgical Teams: A surgical team exhibits collective agency: the surgeon, anesthesiologist, and nurses operate with distinct individual models but a shared awareness of patient state. The team's capability (safe surgery) emerges from their coordination, not from any individual's solo ability. Research shows that surgical outcomes depend more on how well the team works together than on any individual surgeon's skill.

2. Distributed Cognition

No individual can hold the entire organizational model. Collective intelligence distributes the inference task:

- **Perceptual distribution**: Different members attend to different aspects of the environment
- **Cognitive distribution**: Different members maintain expertise in different model components

- **Memory distribution**: Knowledge is stored across people, documents, and tools

  Edwin Hutchins' insight: The cockpit of an airliner "remembers" its speed — not in any pilot's head, but distributed across instruments, checklists, and crew communication protocols. The system remembers, not the individual.

3. Conditions for Collective Intelligence

Research (Woolley et al., 2010) shows that collective intelligence depends on:

Factor	Mechanism	Active Inference Translation
Social sensitivity	Members can infer each other's mental states	Accurate inference about other agents' generative models
Equal participation	No single person dominates the conversation	Distributed sensory sampling across all available channels
Cognitive diversity	Members bring different mental models	Broader coverage of the hypothesis space
Communication quality	Clear, efficient information sharing	High-precision model alignment signals

4. Pathologies of Collective Agency

Pathology	Description	Active Inference Mechanism
Social loafing	Individuals reduce effort in groups	Individual agents decrease active inference when they believe others will compensate
Groupthink	Group suppresses dissenting views	Excessive precision on consensus prior, low precision on individual dissent
Coordination loss	Effort wasted on coordination overhead	Model-alignment costs exceed the benefits of distributed inference
Knowledge hiding	Members withhold information strategically	Individual agents treat information as a competitive resource
Status hierarchy effects	High-status members' views dominate	Precision weighting based on status rather than evidence quality

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Applications

Collective Intelligence Design Checklist

For a team you lead or participate in:

Design Element	Current State	Optimal State	Gap
Social sensitivity			
Participation equality			
Cognitive diversity			
Communication protocols			
Shared representations			
Role clarity			

Cross-References

- For the organization as agent, see Organizational Systems: Agents
- For competitive agents and stakeholders, see Strategic Modeling: Agents
- For AI agents in teams, see Digital Transformation: Agents

Summary

Concept	Team/Collective Meaning
Collective agency	Emergent property of coordinated individual inference
Distributed cognition	Inference task distributed across people, tools, and documents
Role specialization	Each member handles a subset of the collective inference task
Collective intelligence	Group capability that exceeds individual capability
Pathologies	Social loafing, groupthink, coordination loss, status effects

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References

- Woolley, A. W., et al. (2010). Evidence for a collective intelligence factor. *Science*, 330(6004), 686–688.
- Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.
- Edmondson, A. C. (1999). Psychological safety and learning behavior in work teams. *Administrative Science Quarterly*, 44(2), 350–383.